MAS1004 Data & AI, 2026 Fall

Assignment 1: Short report

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Short Essay

Since the AI boom, I've wondered how it could benefit dancers in practical ways without replacing creativity or relying on generative AI, since I don't feel comfortable using it as an artist in general. I first thought about injury prevention and body alignment, but those ideas were too ambitious for this project, as it was my first time trying something like this, and when I first tried the data collection, I instantly saw it was going to be too difficult. Instead, I chose shoe soles because I often struggle to know which shoes are suitable for different dance styles and the best option for me to buy. I thought a simple classifier could be useful for dancers and possibly other athletes like runners.

At the beginning, the project felt overwhelming, and I struggled to understand the workflow and the steps I had to take. However, once I started training and testing the model, I became more invested in the project itself and in how it worked. The most interesting part was experimenting with different image manipulations. The random 4×4 shuffle was the most surprising because all four Nike images were classified as Converse. I also had fun using the drawing tool to test simple shapes and getting the expected results, such as a spiral for Nike, a hexagon for Vans, and a diamond for Converse. During these experiments, I realized that the model probably relies on specific visual patterns rather than recognizing an entire shoe the way we as humans would. I enjoyed this process because it felt like detective work, trying different things to understand the results.

Another important insight I got was how difficult it can be to collect and clean useful data. Before this assignment, I hadn' t thought much about the amount of work behind training an AI model. I knew a bunch of data was needed, but that was mostly it. Going through the process helped me understand why companies need it so much and why it is so valuable. It also made me think if the people' s data that was used to train these models received enough recognition or was given that value it deserves. Although I know that is not the case for art or music, due to all the lawsuits I have seen due to copyright but I still wonder about the rest. This is still a challenging subject for me, but this project helped me understand some basic processes and showed me that I can learn by experimenting, even when I don't understand everything at first. I feel a bit more prepared for Assignment 2 and learned on my mistakes, like relying on free tools instead of just using Claude. I would also be more careful with the data-cleaning process, but I know it got tricky because I had to switch tools halfway through the project.

Screenshots of the model

- The screenshots below are some of my different interactions with the model. Trying out all different ways: webcam, images, and drawing.

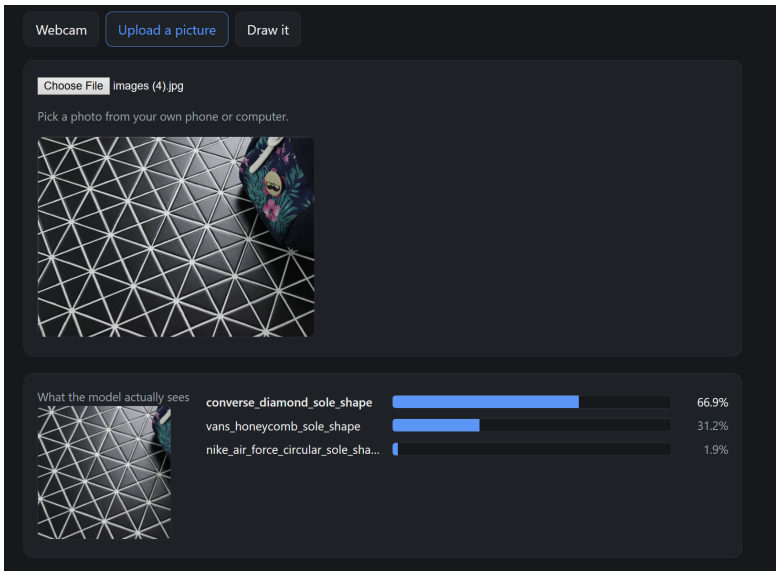


Figure 1: Trying similar shapes as the ones in the Converse sole

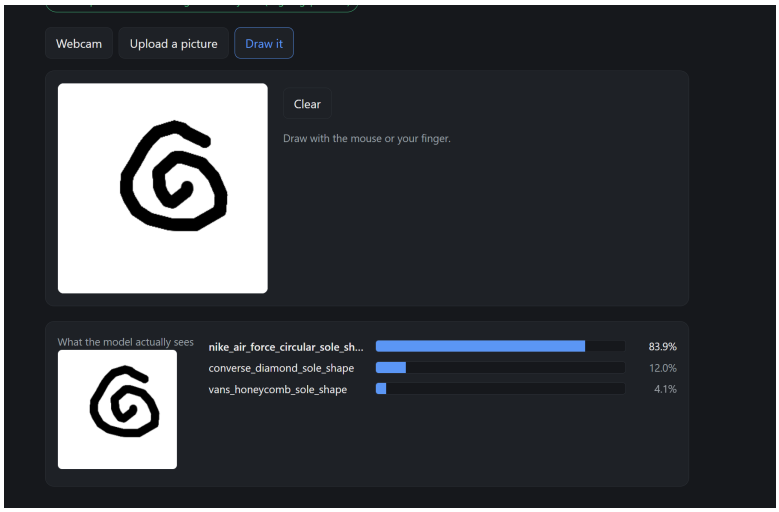


Figure 2: Model identifies a similar drawn shape for Nike soles

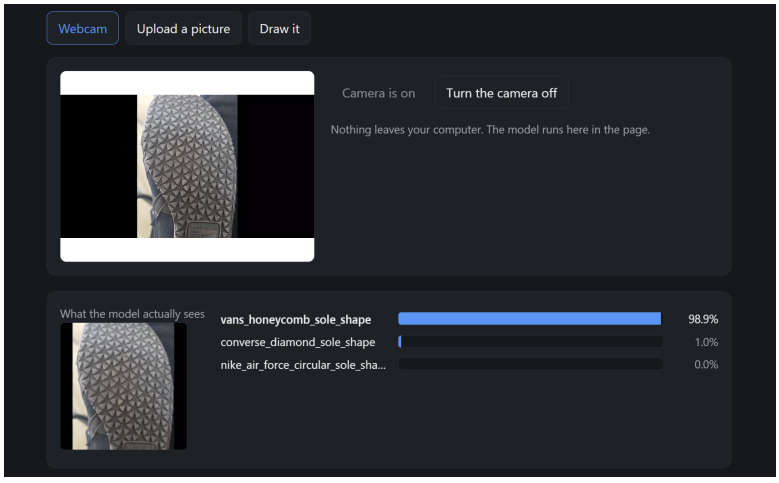


Figure 3: A similar sole to Vans, shape-wise, is correctly classified using the webcam